Lab Log Book

WEEK - 6(Jagadeesh)

Code :

```
[401]: # Student ID (SID = 2450489)
start_point = 50489   # last 5 digits of SID
period = 489          # last 3 digits of SID

# Extract subset for plotting
data_plot = data2.iloc[start_point : start_point + period, :]

# Plot High_Bid and Low_Bid using iplot
data_plot[['High_Bid', 'Low_Bid']].iplot(
    kind='line',
    xTitle='Time (minutes)',
    yTitle='Price',
    title='GOLD: High_Bid and Low_Bid Price Chart (SID: 2450489)',
    colors=['green', 'red']
)
```

Explanation of each step:

1. start_point and period determine the specific segment of the dataset unique to each student.
2. data_plot extracts only those rows from the normalized dataset (data2).
3. iplot() is used for interactive visualization, plotting High_Bid (green line) and Low_Bid (red line).
4. Axis labels and title were added for clarity, and colors were chosen to contrast high vs low price levels.

Output:

Graph Interpretation

- The green line (High_Bid) represents the maximum bid price recorded in each minute, while the red line (Low_Bid) shows the minimum bid price.
- Both lines move in parallel, indicating that market prices fluctuated within a relatively narrow band, suggesting stable trading conditions with mild volatility.
- Occasional spikes and dips indicate short bursts of trading activity or changes in liquidity.
- The price range during this period was roughly between 0.620 and 0.635, showing minor fluctuations over the 489-minute span.

This pattern is typical of intraday financial data, where gold bid prices vary within a small range due to high-frequency trading and global market influences.